

Chapter 8, Problem 3PP

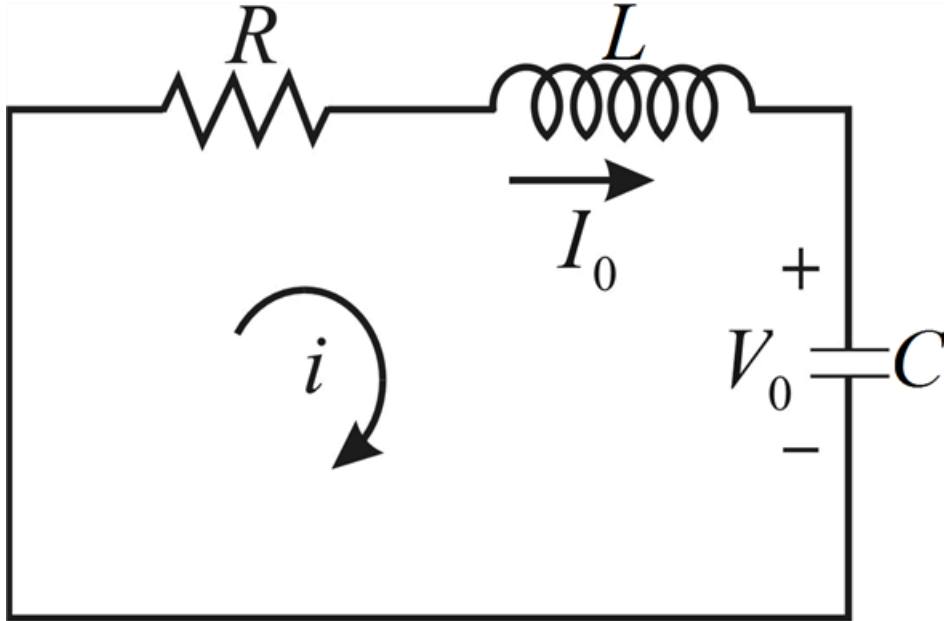
Problem

If $R = 10\ \Omega$, $L = 5\ \text{H}$, and $C = 2\ \text{mF}$ in Fig. 8.8, find α , ω_0 , s_1 , and s_2 . What type of natural response will the circuit have?

Step-by-step solution

Step 1 of 4

The source - free series RLC circuit is



Step 2 of 4

Given

$$R = 10\ \Omega$$

$$L = 5\ \text{H}$$

$$C = 2\ \text{mF}$$

We first calculate

$$\alpha = \frac{R}{2L}$$

$$\alpha = \frac{10}{2 \times 5}$$

$$\therefore \boxed{\alpha = 1}$$

Step 3 of 4

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \frac{1}{\sqrt{5 \times 2 \times 10^{-3}}}$$

$$\omega_0 = \frac{1}{0.1}$$

$$\therefore \boxed{\omega_0 = 10}$$

Step 4 of 4

The roots are

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$s_{1,2} = -1 \pm \sqrt{(1)^2 - (10)^2}$$

$$s_{1,2} = -1 \pm \sqrt{-99}$$

$$s_{1,2} = -1 \pm j9.95$$

$$\therefore \boxed{s_1 = -1 + j9.95}$$

$$\boxed{s_2 = -1 - j9.95}$$

Since $\omega_0 > \alpha$, we conclude that the response is **underdamped**. This is also evident from the fact that the roots are complex.